Midterm - March 17, 2024

Duration: 80 minutes.
Clearly justify all your answers and all the steps taken to explain your reasoning.

Problem 1 (40 points)

Consider the symmetric positive definite matrix A:

$$A = \begin{bmatrix} 4 & -1 & 1 \\ -1 & 4.25 & 2.75 \\ 1 & 2.75 & 3.5 \end{bmatrix}.$$

1. (20) Using a Cholesky-like algorithm (not Gauss elimination), find the LDL^T factorization of A, where L is a lower triangular matrix with 1 in its diagonal and D is a diagonal matrix. Show your work.
2. (10) Calculate the LU factorization of A without using Gauss elimination. Justify.
3. (10) In the Gauss elimination of A, would we need to use any pivoting strategy? Why or why not?

Problem 2 (40 points)

Consider the system $Ax = b$, where A is a non-singular matrix with non-zero diagonal entries. We would like to solve this system with the following iterative method:

$$\begin{cases} (D-E)\, x^{\left(k+\frac{1}{2}\right)} & = Fx^{(k)} + b \\ x^{(k+1)} & = wx^{\left(k+\frac{1}{2}\right)} + (1-w)\, x^{(k)}, \end{cases}$$

in which w is a real parameter, D is the diagonal part of A, E is the lower triangular part of $-A$ (without the diagonal) and F is the upper triangular part of $-A$ (without the diagonal).

1. (15) Rewrite the iterative method in the form

$$x^{(k+1)} = B_w x^{(k)} + f_w,$$

showing B_w and f_w explicitly. Justify your steps.

2. (5) Do you recognize this method when $w = 1$? If so, what is the name of the method?

3. Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix} \qquad \text{and} \qquad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

   a) (10) Find the first iterate if $x^{(0)} = [1 \quad 1]^T$ and any w.

   b) (10) What are the values of w to have convergence of the method? Justify.

Problem 3 (20 points)

Suppose we wish to determine the spring constant k in Hooke's law,

$$F(l) = k(l - E),$$

in which the function F is the force (lb) required to stretch the spring l units (in) and $E = 5.3$ in is the natural spring's length. Consider the data given in the following table:

$F(l)$ (lb)	l (in)
2	7.0
4	9.4
6	12.3

Find the least squares approximation for k and the corresponding RMS error. Show and justify your work.